

Midterm review and the evidence audit

Can we find a claim that outruns the data?

By the end, you should be able to...



Retrieve the foundational skills from Weeks 1-7.



Classify a statement as supported, contradicted or not established.



Replace an overclaim with a bounded scientific conclusion.

Reconnect with the biology

The midterm tests the floor: biology-to-data reasoning, coordinates, short code, depth and statistical interpretation.

Think silently -> explain to a neighbour -> check the measurement.

The midterm is about reasoning

01

60 minutes; structured questions; a supplied formula sheet.

>

02

Show units, steps and a limited interpretation.

>

03

No internet or AI during the paper; approved accommodations still apply.

Retrieve: same interval, different notation

Same bases. Different ruler.

1	2	3	4	5	6	7	8	9	10
A	C	G	T	A	C	G	T	A	A
0	1	2	3	4	5	6	7	8	9

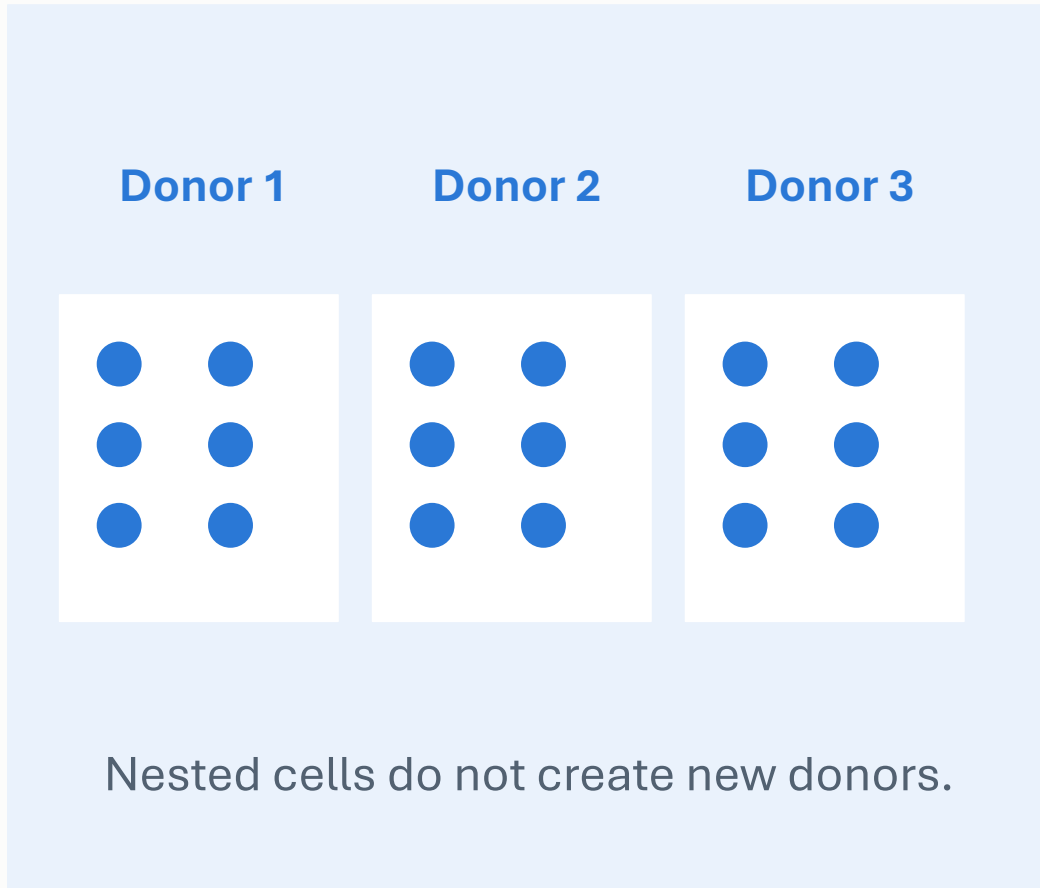
1-based positions above / 0-based indices below

BED [2, 7) has length 5.

One-based inclusive [3, 7] describes the same bases.

A missing assembly must be reported as unknown.

Retrieve: count the independent origin



Two donors with 200 cells each are two donors.

A replicate label must name the biological unit.

More sequencing cannot create a missing biological replicate.

Retrieve: normalize for the intended task

01

Do not confuse
observed counts with
relative abundance.

>

02

The denominator must
match the calculation's
definition.

>

03

A successful run is not
proof of an appropriate
analysis.

Three evidence labels

SUPPORTED: the supplied evidence supports this exact scope.

CONTRADICTED: the evidence or definition conflicts with the claim.

NOT ESTABLISHED: additional evidence is required.

Audit the statement, not the writing style

A fluent paragraph may mix correct facts with an invalid inference.

Underline each factual claim and identify its evidence.

Use tests and sources; do not use an AI-text detector as proof.

Worked audit: high counts, strong claim

Claim: Gene X has twice as many RNA reads in one treated sample, so the treatment doubles protein activity.

Predict the result and one limitation before calculating.

Work through the reasoning

The raw-count comparison is supplied evidence.

The treatment effect lacks independent replication and normalization context.

Protein activity was not measured; that conclusion is not established.

Pause: make a defensible claim

A report says: 1,000 cells from one donor prove the result generalizes to all adults. Assign an evidence label and propose a better sentence.

First answer alone. Then compare the evidence, not just the conclusion.

A defensible response

NOT ESTABLISHED. A better sentence is: this donor's measured cells show the reported pattern; generalization requires independent donors and an appropriate design.

Today's practical

W08: audit six supplied statements after the midterm, attach evidence and correct two overclaims. The audit is a completion lab, not an extra exam.

Minutes	Action
0-10	Predict + set up
10-30	Run + inspect + check
30-40	Explain + compare
40-45	Save + submit / exit

Five terms to use precisely

1 evidence

2 scope

3 control

4 contradiction

5 limitation

Exit ticket

What did we actually measure or calculate?

What is one assumption or limitation of that result?

What check would you perform before sharing it?

Go further only after the core is clear

Optional: exchange one evidence audit with a peer and check whether you agree on the scope.

Not required for the core grade. No advanced model training is expected.

Read the evidence, not only the slide

S03 SAM/BAM, VCF and BED specifications, maintained by the HTS community.

S04 UCSC Genome Browser: data format FAQ.

S07 American Statistical Association. Statement on statistical significance and p-values (2016).

S10 DESeq2: Analyzing RNA-seq data with DESeq2, release vignette.

S12 Squair et al. (2021). Confronting false discoveries in single-cell differential expression.

S23 ICMJE recommendations: use of AI by authors.